

# Cluster Size vs $E_T$ : Data vs MC, Split vs No-Split, Single vs Double Interaction

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## 1 Motivation

Cluster size (the number of EMCal towers associated with a cluster and the transverse extent of that set of towers) is a basic shape property that underlies every shower-shape-based background discriminant used in PPG12 (NPB, shower-shape BDT, width variables). This report inspects how that size depends on cluster  $E_T$  across three orthogonal handles:

- **Sample:** data, signal MC (isolated prompt photons), background MC (QCD jets).
- **Pileup:** single-interaction MC, full-GEANT double-interaction MC, and the physics-weighted mix at 7.9% (1.5 mrad crossing angle) and 22.4% (0 mrad). Only MC is split this way; data is shown as a single curve against the two mixed references.
- **Clusterizer:** the nominal split-cluster container `CLUSTERINFO_CEMC` versus the legacy non-split `CLUSTERINFO_CEMC_NO_SPLIT`. Both containers are stored in every slimtree, so the comparison is an apples-to-apples rerun of the two CEMC clusterizer modes on the same events.

## 2 Definitions

For each cluster  $c$  we read the  $7 \times 7$  tower matrix `cluster_ownership_arrayc` and extract the set of towers marked as owned by the cluster (value 1). Let  $\mathcal{O}_c$  be the owned-tower index set, with  $|\mathcal{O}_c| = n_{\text{owned}}$ . Each index  $i \in [0, 48]$  corresponds to a position in the  $7 \times 7$  grid with  $i_\eta = \lfloor i/7 \rfloor$ ,  $i_\phi = i \bmod 7$ , the center tower at  $(3, 3)$  (see `CaloAna24.cc:1403`).

- $n_{\text{owned}}$ :  $|\mathcal{O}_c|$  — literal cluster tower count
- $\Delta i_\eta$ :  $\max_{i \in \mathcal{O}_c} i_\eta - \min_{i \in \mathcal{O}_c} i_\eta + 1$  — bounding-box extent in  $\eta$
- $\Delta i_\phi$ : analogous in  $\phi$

Both `CLUSTERINFO_CEMC` and `CLUSTERINFO_CEMC_NO_SPLIT` are computed identically. No shower-shape, BDT, NPB, or isolation cut is applied; the only fiducial requirement is  $|\eta_{\text{reco}}| < 0.7$ .

### 3 Samples

| Category          | Samples                                   | Entries (event)                      | Notes                                            |
|-------------------|-------------------------------------------|--------------------------------------|--------------------------------------------------|
| Data              | 77 <code>part*_with_bdt_split.root</code> | DATA_NEVT                            | run 47289–54000 (both crossing angles)           |
| Signal single     | photon5, photon10, photon20               | stitched by truth $p_T^\gamma$       | $\sigma_{\text{ref}} = \sigma_{\text{photon20}}$ |
| Signal double     | photon10_double                           | $p_T^\gamma \in [10, 100]$           | full-GEANT double interaction                    |
| Background single | jet5, jet8, jet12, jet20, jet30, jet40    | stitched by truth $p_T^{\text{jet}}$ | $\sigma_{\text{ref}} = \sigma_{\text{jet50}}$    |
| Background double | jet12_double                              | $p_T^{\text{jet}} \in [10, 100]$     | full-GEANT double interaction                    |

Table 1: Sample inventory. Truth  $p_T$  windows (from `CrossSectionWeights.h`) are applied in the analysis macro to avoid double counting across overlapping generator windows.

Per-event cross-section weights are taken from `PPG12::GetSampleConfig` (relative to each class’s reference sample). Cluster-level mixing uses a bin-by-bin linear combination of profile means:

$$\langle y \rangle_{\text{mix}}(E_T) = (1 - f) \langle y \rangle_{\text{single}}(E_T) + f \langle y \rangle_{\text{double}}(E_T), \quad (1)$$

with  $f = 0.079$  (1.5 mrad) or  $f = 0.224$  (0 mrad). Note that this is a linear combination of the per-bin means, not a weighted merger of underlying fills — the latter would be biased by the relative event counts of the single and double samples.

### 4 Results

## 4.1 Nominal data vs inclusive MC

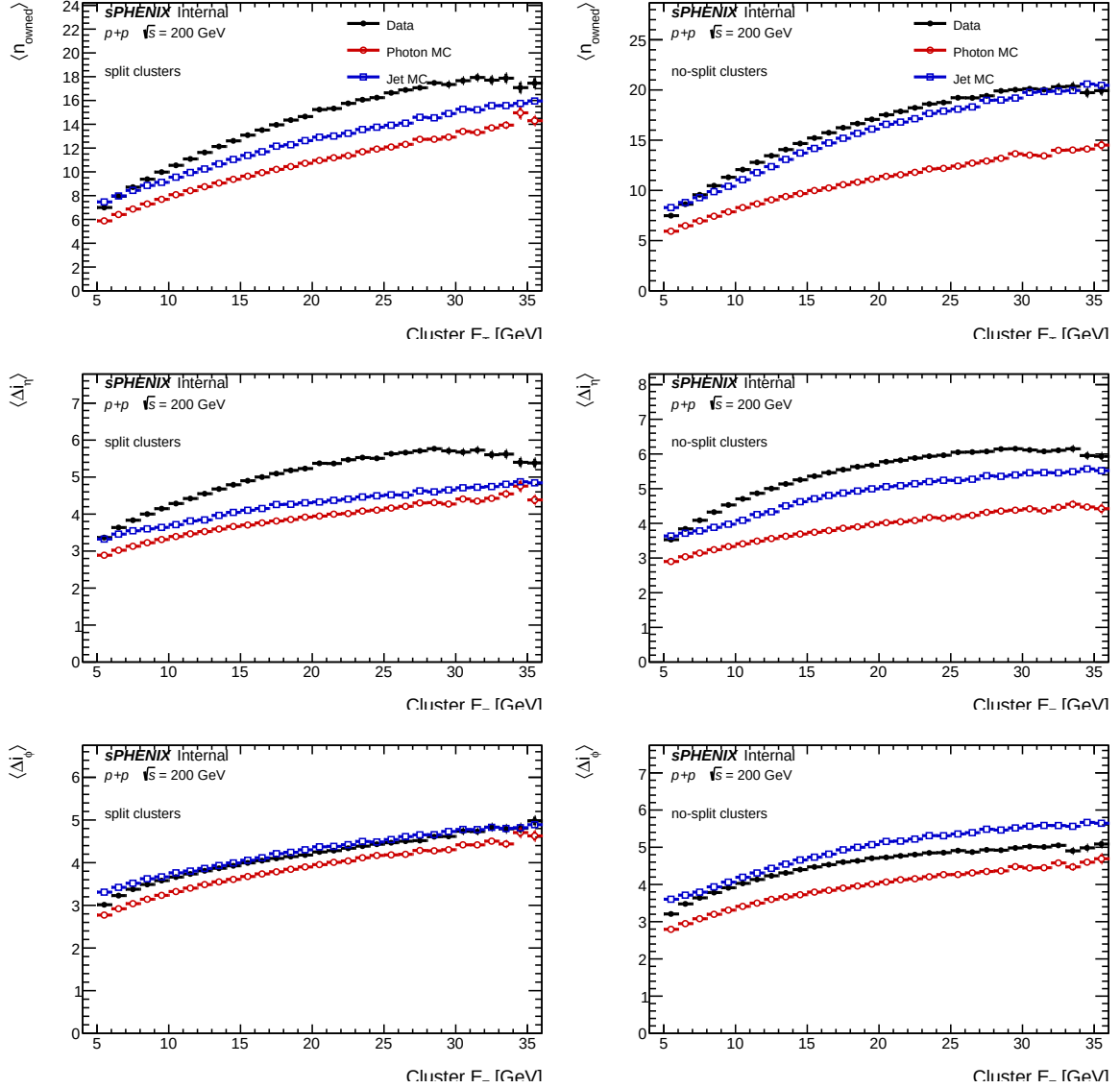


Figure 1: Mean cluster size vs  $E_T$  for data (black), pooled signal MC (photon5+10+20, red) and pooled background MC (jet5+8+12+20+30+40, blue), with MC weights from GetSampleConfig. Rows:  $n_{\text{owned}}$ ,  $\Delta i_\eta$ ,  $\Delta i_\phi$ . Columns: split (CLUSTERINFO\_CEMC), no-split (CLUSTERINFO\_CEMC\_NO\_SPLIT). Only  $|\eta_{\text{reco}}| < 0.7$  is applied.

### Observations.

- In the split container, data is systematically larger than both MC curves for  $n_{\text{owned}}$  and  $\Delta i_\eta$ . At  $E_T = 20$  GeV, data has  $\langle n_{\text{owned}} \rangle = 15.2$ , jet MC has 12.9, photon MC has 11.0 — a data–jet-MC gap of +2.3 towers ( $\sim 18\%$ ). The gap grows from  $\approx 0$  at  $E_T = 5$  GeV to a peak of +2.7 at  $E_T = 25$  GeV.
- The  $\Delta i_\phi$  panel is the **exception**: data and jet MC agree within a few percent across the full range. Only the  $\eta$  direction is under-modelled. This asymmetry is consistent with the extra towers being preferentially stretched in  $\eta$  — possibly electronic noise or pedestal drifts on isolated towers that the clusterizer then absorbs into a nearby cluster, plus the larger physical tower pitch in  $\eta$  favouring one-tower spillovers.

- **The data–MC gap is much smaller in the no-split container:**  $\approx +1.0$  tower across  $E_T = 10\text{--}25$  GeV, shrinking to  $+0.4$  at  $E_T = 30$  GeV (essentially converged). The extra towers in data *are* picked up by the no-split clusterizer but not fragmented off by the split one — consistent with a diffuse, noise-like contribution spread across the  $7\times 7$  matrix rather than a separable sub-cluster. If these were genuine secondary showers, the split algorithm would have removed them, leaving the data–MC gap the same in both containers.
- Jet clusters (blue) are larger than photon clusters (red) across the full  $E_T$  range in every metric, as expected — jets deposit energy over multiple overlapping subshowers while prompt photons produce a compact EM shower. The gap widens in `CLUSTERINFO_CEMC_NO_SPLIT` because splitting preferentially shrinks jet clusters while leaving photons alone (quantified in Fig. 4).
- Both MC populations rise with  $E_T$  because the 70 MeV tower threshold becomes easier to exceed as the total shower energy grows. Data shows the same trend with a slightly steeper slope — the data–jet-MC gap in the split container grows from essentially zero at  $E_T = 5$  GeV to  $\sim 2.7$  towers at  $E_T = 25$  GeV, then starts levelling off.

## 4.2 Single vs double interaction — photon signal

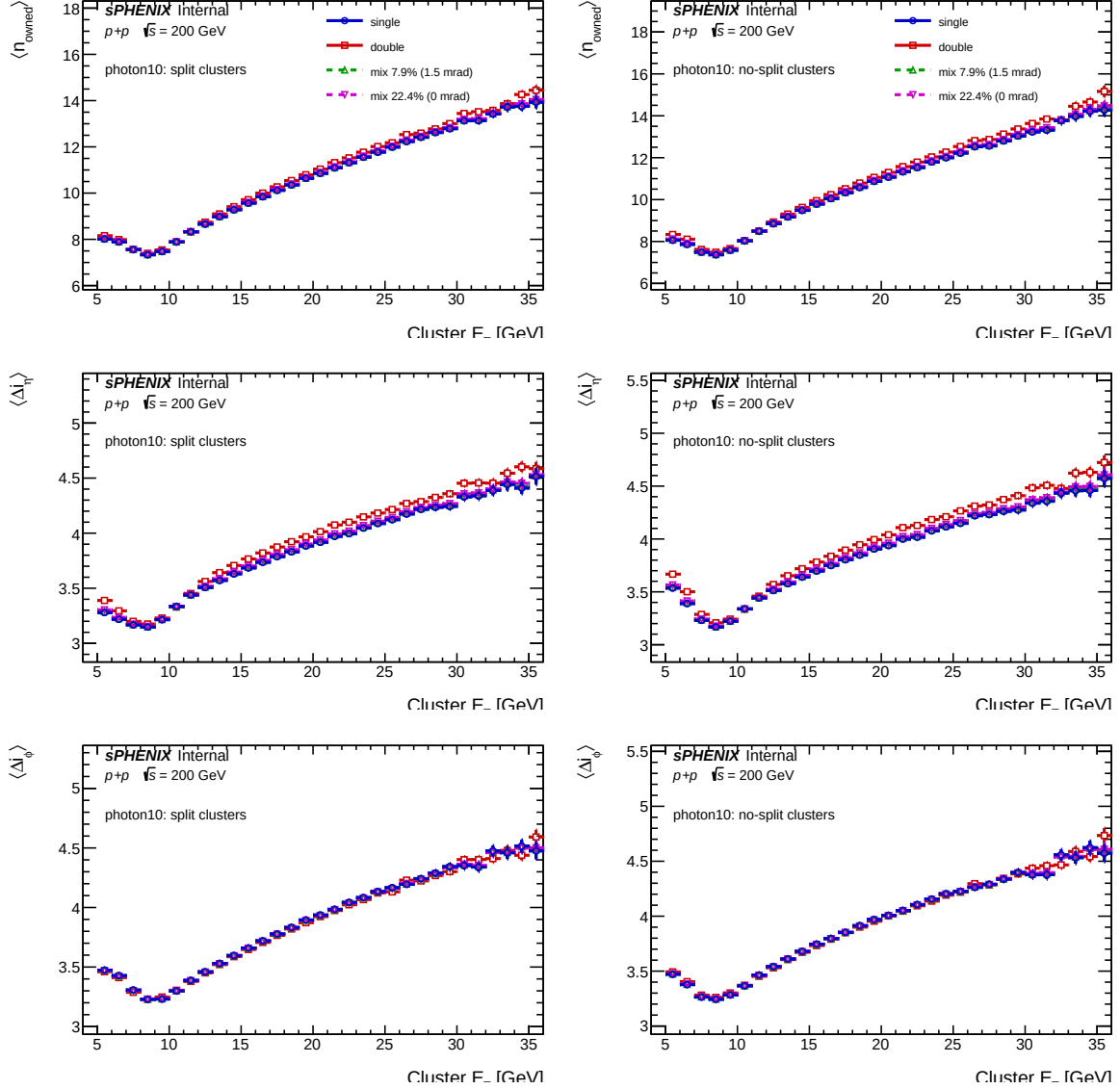


Figure 2: Photon MC cluster size vs  $E_T$ : single interaction (photon10\_nom, blue), pure double interaction (photon10\_double, red), and linearly mixed at 7.9% (green, 1.5 mrad) and 22.4% (magenta, 0 mrad). Mixing applied per Eq. (1).

### Observations.

- Double interaction shifts photon cluster size *upwards* by  $\Delta \langle n_{\text{owned}} \rangle \approx +0.15$  to  $+0.22$  towers above single across  $E_T \in [15, 30]$  GeV. The effect is real but small — pile-up adds a few soft neighbours to an otherwise compact EM shower.
- The bounding-box widths ( $\Delta i_\eta$ ,  $\Delta i_\phi$ ) shift by  $\lesssim 0.1$  between single and double; the 1<sup>st</sup>-order effect of double interaction on photon shower width is at the sub-0.1 index level.
- Mixed curves at 7.9% (1.5 mrad) and 22.4% (0 mrad) sit between single and pure-DI at the corresponding fractions. The mixed-7.9% curve is nearly indistinguishable from single — consistent with the shower-shape BDT being pile-up-robust in the 1.5 mrad regime.
- The dip near  $E_T = 9$  GeV is the energy-resolution tail of the  $p_T^{\gamma, \text{truth}} > 10$  generator window; below-threshold reco clusters here are dominated by leakage, not peak showers.

- The split vs no-split containers behave almost identically for photon MC (see Fig. 4), so the DI effect is also essentially independent of the clusterizer.

### 4.3 Single vs double interaction — jet background

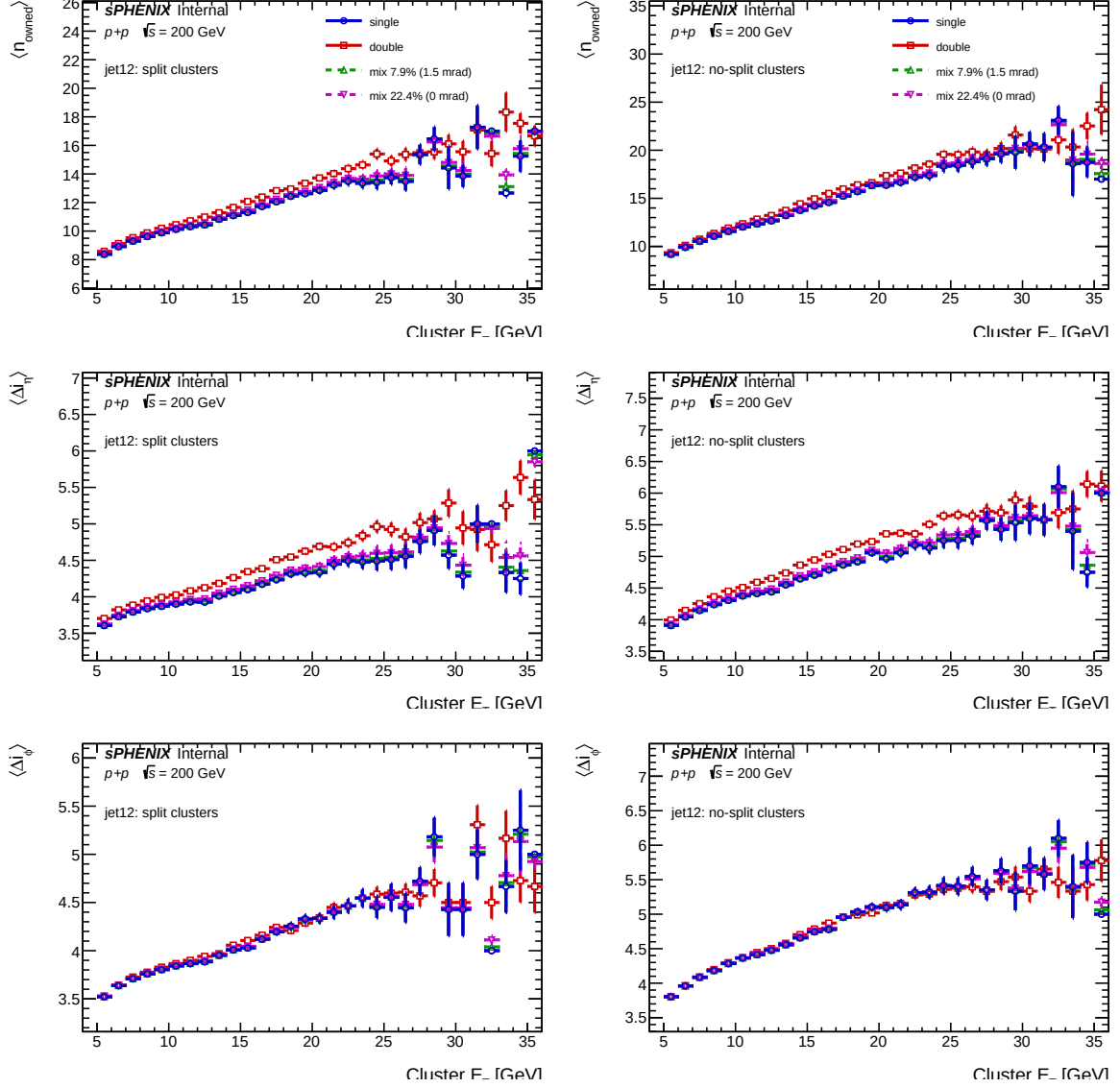


Figure 3: Jet MC cluster size vs  $E_T$ , same format as Fig. 2 but for `jet12_nom` vs `jet12_double`.

#### Observations.

- Double interaction inflates *jet* cluster size much more strongly than photon cluster size:  $\Delta\langle n_{\text{owned}} \rangle \approx +0.5$  to  $+1.3$  towers in the split container at  $E_T > 15$  GeV, roughly  $4\times$  larger than the photon response.
- Physically: a jet cluster already has multiple soft neighbouring towers from accompanying hadrons, and DI adds more of the same; the probability that DI overlay lands on a low- $E$  tower and pushes it over the 70 MeV threshold is much higher in a jet environment than for a clean photon.
- The DI-induced broadening is partially absorbed by the no-split container (right column), which already inflates jet clusters via overlap, so the relative DI effect is smaller there.

- At  $E_T > 30$  GeV the **jet12** sample runs out of statistics (truth  $p_T^{\text{jet}} \in [14, 21]$  GeV window), producing the noisy tail seen in every jet12 DI panel.
- Because jets enter both the fake-photon and anti-tight control regions of the ABCD, this DI effect is a non-trivial input to the background-template closure: in 0 mrad running (22.4% DI), the jet template mean cluster size shifts upward by  $\sim 0.3$ – $0.4$  towers, which affects the shower-shape BDT input distribution and therefore the background rejection rate.

#### 4.4 No-split / split ratio

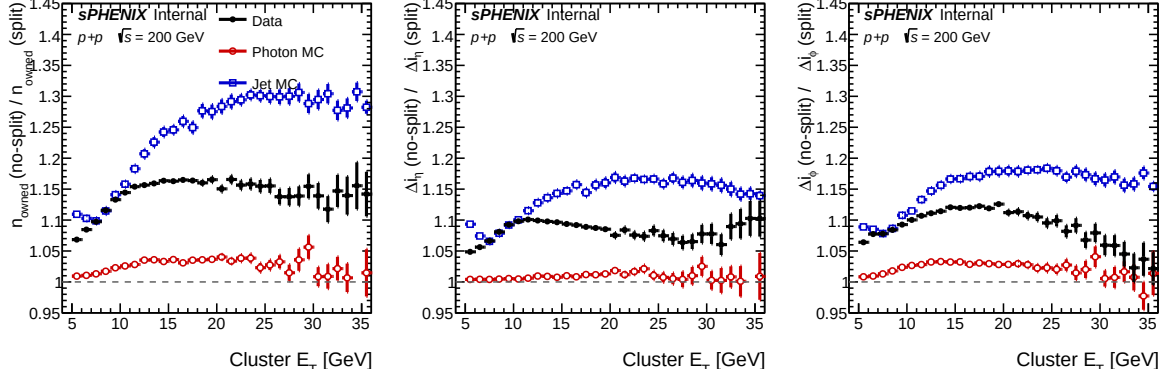


Figure 4: Ratio of mean cluster size in the no-split container to that in the split container, vs  $E_T$ . Solid black: data. Red: pooled photon MC. Blue: pooled jet MC. A flat ratio of 1 would mean the two clusterizer modes are indistinguishable.

#### Observations.

- **Photon MC (red)** has a nearly flat ratio of 1.01–1.04 in every panel: splitting has essentially no effect on clean single-photon clusters.
- **Jet MC (blue)** shows a clear  $E_T$ -dependent effect:  $n_{\text{owned}}$  ratio rises from  $\approx 1.10$  at  $E_T = 7$  GeV to  $\approx 1.30$  at  $E_T \gtrsim 20$  GeV. The bounding-box widths ( $\Delta i_\eta$ ,  $\Delta i_\phi$ ) show the same trend at a milder level, saturating around 1.16–1.18. Splitting removes  $\sim 30\%$  of jet cluster towers at  $E_T > 15$  GeV.
- **Data (black)** sits between photon MC and jet MC, with a split/no-split ratio of  $\approx 1.10$ – $1.17$  for  $n_{\text{owned}}$  — below the jet-MC expectation of 1.30. This means the extra towers that data has compared to MC (Sec. 4.1) are *not* being removed by the split clusterizer — they are distributed diffusely, not as separable sub-clumps. Physically consistent with noise/threshold contamination rather than genuine secondary showers.
- The data  $n_{\text{owned}}$  ratio peaks near  $E_T = 15$ – $20$  GeV and declines slightly at higher  $E_T$ , whereas jet MC stays flat or gently rises. Data is approaching the 49-tower matrix cap in CLUSTERINFO\_CEMC\_NO\_SPLIT at  $E_T \gtrsim 30$  GeV, so the ratio numerator saturates before the denominator.

## 5 Summary

Four findings matter for the PPG12 analysis:

1. **Data is systematically larger than jet MC in the split container** in  $\langle n_{\text{owned}} \rangle$  and  $\langle \Delta i_\eta \rangle$  by +1 to +2.7 towers ( $\sim 10$ – $20\%$  of the mean) across  $E_T = 10$ – $30$  GeV. The data–MC gap is  $\sim 3\times$  smaller in the no-split container.  $\langle \Delta i_\phi \rangle$  data-MC agreement is *much* better. This is the

single biggest finding: the clusterizer MC does not reproduce the tower-multiplicity of real data. Most plausible causes are noise-threshold miscalibration or un-modelled underlying-event / zero-bias contamination; electronic noise is the likeliest root cause given the  $\eta$ -only asymmetry and the absence of a matching shift in  $\phi$ .

2. **The extra data-MC towers are diffuse, not clumped.** In the no-split/split ratio (Fig. 4), data sits *below* jet MC: extra towers in data do not form separable sub-clusters that the split algorithm could fragment off, so splitting has less to remove. The implication for the PPG12 shower-shape BDT is that it should be trained and validated against data-like (not MC-like) cluster morphology to avoid over-tightening at high  $E_T$ .
3. **Cluster splitting is a photon-safe jet shrinker.** The split clusterizer removes  $\sim 30\%$  of jet cluster towers at  $E_T > 15$  GeV while leaving photon cluster size unchanged to within a few percent (red curve in Fig. 4). This is the physical mechanism by which CLUSTERINFO\_CEMC improves signal-background shape separation and motivates the nominal pipeline’s choice of the split container.
4. **Double interaction broadens jet clusters much more than photon clusters.** Pure DI adds  $\sim 0.5$ – $1.3$  towers to jet cluster size (4–10% of the mean) and only  $\sim 0.2$  towers to photon cluster size (2% of the mean). At 1.5 mrad running (7.9% DI), the effect on the weighted mean is absorbed by systematics, but at 0 mrad (22.4% DI) the jet-cluster shift is a concrete mechanism for the DI systematic on the background template shape. Photon DI is sub-leading.

## Follow-ups suggested by this study

- **Investigate the data-MC cluster-size gap.** Cross-check by running the same macro on a minbias-triggered data subset, with and without a  $|t_{\text{MBD}}| < 25$  ns cut, to isolate whether the gap is from noise, underlying event, or pile-up.
- Compare the noise-tower distributions between data and MC (per-channel RMS, hit rates above the 70 MeV threshold in empty fiducial regions) to quantify the noise-contamination hypothesis.
- Once the noise/threshold question is settled, decide whether the BDT training sample should include a data-like noise overlay.

## Reproducibility

Generation:

```
cd efficiencytool && bash run_cluster_size.sh
cd ../plotting && root -l -b -q plot_cluster_size.C+
```

Key macros:

- `efficiencytool/ClusterSizeStudy.C` — computes  $n_{\text{owned}}$ ,  $\Delta i_\eta$ ,  $\Delta i_\phi$  per cluster, both cluster nodes, with truth- $p_T$  sample stitching and fiducial  $|\eta_{\text{reco}}| < 0.7$ .
- `efficiencytool/run_cluster_size.sh` — driver (6-way parallel).
- `plotting/plot_cluster_size.C` — figures.